



ANDROID STATIC ANALYSIS REPORT

app_icon

 Aegis (1.0)

File Name:	app-debug.apk
Package Name:	com.example.projectaegis
Scan Date:	July 20, 2026, 9:16 p.m.
App Security Score:	38/100 (HIGH RISK)

Grade:



 FINDINGS SEVERITY

 HIGH	 MEDIUM	 INFO	 SECURE	 HOTSPOT
3	3	2	1	0

 FILE INFORMATION

File Name: app-debug.apk
Size: 25.42MB
MD5: b157f7a15b60d11bc941ee2adcfb9d39
SHA1: 4710985eb939313b68e00a8c4b3b615ebb3af871
SHA256: 3469e65fe9e558d035348d75495879efa0c2bfc005ed52ce0beef323419174bb

 APP INFORMATION

App Name: Aegis
Package Name: com.example.projectaegis
Main Activity: com.example.projectaegis.MainActivity
Target SDK: 36
Min SDK: 24
Max SDK:
Android Version Name: 1.0
Android Version Code: 1

 APP COMPONENTS

Activities: 5
Services: 0
Receivers: 1
Providers: 1
Exported Activities: 0
Exported Services: 0
Exported Receivers: 1
Exported Providers: 0

CERTIFICATE INFORMATION

Binary is signed
v1 signature: False
v2 signature: True
v3 signature: False
v4 signature: False
X.509 Subject: CN=Android Debug, O=Android, C=US
Signature Algorithm: rsassa_pkcs1v15
Valid From: 2026-07-06 16:03:40+00:00
Valid To: 2056-06-28 16:03:40+00:00
Issuer: CN=Android Debug, O=Android, C=US
Serial Number: 0x1
Hash Algorithm: sha256
md5: 80825ee1b17cf23de0a560f005e46c43
sha1: 92d74402a55977402ad29646ad4d5685bcf96a7c
sha256: f37e31c34641c6c5ebe5a71e85a8fb71fe7914d89ff832edbc59e22ae923c522
sha512: c42cf8e6c8591ad076a7fe985c406bf5d12d0d6faf3ac5c0f8bb33e2e5ddf31ce327ad48700af63cf31ec4b701d20b55953575b5e31e9d0be6c479b998a47024
PublicKey Algorithm: rsa
Bit Size: 2048
Fingerprint: 6a6514a4d943453575e87da462d84d5e2f11f1f736ad3e7b27f0d87656be37fb
Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.USE_BIOMETRIC	normal	allows use of device-supported biometric modalities.	Allows an app to use device supported biometric modalities.
android.permission.USE_FINGERPRINT	normal	allow use of fingerprint	This constant was deprecated in API level 28. Applications should request USE_BIOMETRIC instead.
com.example.projectaegis.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION	unknown	Unknown permission	Unknown permission from android reference

 APKID ANALYSIS

FILE	DETAILS	
classes.dex	FINDINGS	DETAILS
	Anti-VM Code	Build.FINGERPRINT check Build.MODEL check Build.MANUFACTURER check
	Compiler	r8

FILE	DETAILS	
classes2.dex	FINDINGS	DETAILS
	Compiler	unknown (please file detection issue!)
classes3.dex	FINDINGS	DETAILS
	Compiler	r8
classes4.dex	FINDINGS	DETAILS
	Compiler	r8
classes5.dex	FINDINGS	DETAILS
	Compiler	r8
classes6.dex	FINDINGS	DETAILS
	Compiler	r8

FILE	DETAILS	
classes7.dex	FINDINGS	DETAILS
	Compiler	r8
classes8.dex	FINDINGS	DETAILS
	Compiler	r8 without marker (suspicious)
classes9.dex	FINDINGS	DETAILS
	Compiler	r8 without marker (suspicious)

 NETWORK SECURITY

NO	SCOPE	SEVERITY	DESCRIPTION
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 CERTIFICATE ANALYSIS

HIGH: 1 | WARNING: 0 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate
Application signed with debug certificate	high	Application signed with a debug certificate. Production application must not be shipped with a debug certificate.

MANIFEST ANALYSIS

HIGH: 2 | WARNING: 1 | INFO: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 7.0, [minSdk=24]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Debug Enabled For App [android:debuggable=true]	high	Debugging was enabled on the app which makes it easier for reverse engineers to hook a debugger to it. This allows dumping a stack trace and accessing debugging helper classes.
3	Broadcast Receiver (androidx.profileinstaller.ProfileInstallReceiver) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.DUMP [android:exported=true]	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.

CODE ANALYSIS

HIGH: 0 | WARNING: 2 | INFO: 2 | SECURE: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	The App logs information. Sensitive information should never be logged.	info	CWE: CWE-532: Insertion of Sensitive Information into Log File OWASP MASVS: MSTG-STORAGE-3	net/zetetic/database/LogcatTarget.java net/zetetic/database/sqlcipher/CloseGuard.java net/zetetic/database/sqlcipher/SQLiteDatabase.java
2	App uses SQLite Database and execute raw SQL query. Untrusted user input in raw SQL queries can cause SQL Injection. Also sensitive information should be encrypted and written to the database.	warning	CWE: CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') OWASP Top 10: M7: Client Code Quality	net/zetetic/database/DatabaseUtils.java net/zetetic/database/sqlcipher/SQLiteDatabase.java
3	This App copies data to clipboard. Sensitive data should not be copied to clipboard as other applications can access it.	info	OWASP MASVS: MSTG-STORAGE-10	com/example/projectaegis/util/ClipboardUtil.java
4	Files may contain hardcoded sensitive information like usernames, passwords, keys etc.	warning	CWE: CWE-312: Cleartext Storage of Sensitive Information OWASP Top 10: M9: Reverse Engineering OWASP MASVS: MSTG-STORAGE-14	com/example/projectaegis/data/VaultDbHelper.java com/example/projectaegis/security/VaultKeyProvider.java

SHARED LIBRARY BINARY ANALYSIS

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
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NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
1	arm64-v8a/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__strlen_chk', '__memcpy_chk', '__memset_chk', '__strchr_chk', '__memmove_chk', '__FD_SET_chk', '__read_chk', '__strncpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
2	x86/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__strlen_chk', '__memcpy_chk', '__memset_chk', '__strchr_chk', '__memmove_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
3	armeabi-v7a/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__memcpy_chk', '__strlen_chk', '__memset_chk', '__strchr_chk', '__memmove_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
4	x86_64/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__strlen_chk', '__memcpy_chk', '__memset_chk', '__strchr_chk', '__memmove_chk', '__FD_SET_chk', '__read_chk', '__strncpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
5	arm64-v8a/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__strlen_chk', '__memcpy_chk', '__memset_chk', '__strchr_chk', '__memmove_chk', '__FD_SET_chk', '__read_chk', '__strncpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
6	x86/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__strlen_chk', '__memcpy_chk', '__memset_chk', '__strchr_chk', '__memmove_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
7	armeabi-v7a/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__memcpy_chk', '__strlen_chk', '__memset_chk', '__strchr_chk', '__memmove_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
8	x86_64/libsqlcipher.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['_vsnprintf_chk', '__strlen_chk', '__memcpy_chk', '__memset_chk', '__strchr_chk', '__memmove_chk', '__FD_SET_chk', '__read_chk', '__strncpy_chk']	True info Symbols are stripped.

NIAP ANALYSIS v1.3

NO	IDENTIFIER	REQUIREMENT	FEATURE	DESCRIPTION
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BEHAVIOUR ANALYSIS

RULE ID	BEHAVIOUR	LABEL	FILES
00022	Open a file from given absolute path of the file	file	net/zetetic/database/sqlcipher/SQLiteDatabase.java

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	0/25	
Other Common Permissions	0/44	

Malware Permissions:

Top permissions that are widely abused by known malware.

Other Common Permissions:

Permissions that are commonly abused by known malware.

HARDCODED SECRETS

POSSIBLE SECRETS
4b55be2c9864cfa0f3e2262a2208567ab6bc862a59e7853c580a1f24fbae9ba1

☰ SCAN LOGS

Timestamp	Event	Error
2026-07-20 21:16:42	Generating Hashes	OK
2026-07-20 21:16:42	Extracting APK	OK
2026-07-20 21:16:42	Unzipping	OK
2026-07-20 21:16:43	Parsing APK with androguard	OK
2026-07-20 21:16:44	Extracting APK features using aapt/aapt2	OK
2026-07-20 21:16:45	Getting Hardcoded Certificates/Keystores	OK
2026-07-20 21:16:48	Parsing AndroidManifest.xml	OK
2026-07-20 21:16:48	Extracting Manifest Data	OK
2026-07-20 21:16:48	Manifest Analysis Started	OK

2026-07-20 21:16:48	Performing Static Analysis on: Aegis (com.example.projectaegis)	OK
2026-07-20 21:16:49	Fetching Details from Play Store: com.example.projectaegis	OK
2026-07-20 21:16:49	Checking for Malware Permissions	OK
2026-07-20 21:16:49	Fetching icon path	OK
2026-07-20 21:16:49	Library Binary Analysis Started	OK
2026-07-20 21:16:49	Analyzing apktool_out/lib/arm64-v8a/libsqlcipher.so	OK
2026-07-20 21:16:49	Analyzing apktool_out/lib/x86/libsqlcipher.so	OK
2026-07-20 21:16:49	Analyzing apktool_out/lib/armeabi-v7a/libsqlcipher.so	OK
2026-07-20 21:16:49	Analyzing apktool_out/lib/x86_64/libsqlcipher.so	OK
2026-07-20 21:16:50	Analyzing lib/arm64-v8a/libsqlcipher.so	OK
2026-07-20 21:16:50	Analyzing lib/x86/libsqlcipher.so	OK

2026-07-20 21:16:50	Analyzing lib/armeabi-v7a/libsqlcipher.so	OK
2026-07-20 21:16:50	Analyzing lib/x86_64/libsqlcipher.so	OK
2026-07-20 21:16:50	Reading Code Signing Certificate	OK
2026-07-20 21:16:50	Running APKiD 3.1.0	OK
2026-07-20 21:16:53	Detecting Trackers	OK
2026-07-20 21:16:56	Decompiling APK to Java with JADX	OK
2026-07-20 21:17:15	Converting DEX to Smali	OK
2026-07-20 21:17:15	Code Analysis Started on - java_source	OK
2026-07-20 21:17:17	Android SBOM Analysis Completed	OK
2026-07-20 21:17:23	Android SAST Completed	OK
2026-07-20 21:17:23	Android API Analysis Started	OK

2026-07-20 21:17:28	Android API Analysis Completed	OK
2026-07-20 21:17:29	Android Permission Mapping Started	OK
2026-07-20 21:17:33	Android Permission Mapping Completed	OK
2026-07-20 21:17:33	Android Behaviour Analysis Started	OK
2026-07-20 21:17:37	Android Behaviour Analysis Completed	OK
2026-07-20 21:17:37	Extracting Emails and URLs from Source Code	OK
2026-07-20 21:17:38	Email and URL Extraction Completed	OK
2026-07-20 21:17:38	Extracting String data from APK	OK
2026-07-20 21:17:38	Extracting String data from SO	OK
2026-07-20 21:17:38	Extracting String data from Code	OK
2026-07-20 21:17:38	Extracting String values and entropies from Code	OK

2026-07-20 21:17:40	Performing Malware check on extracted domains	OK
2026-07-20 21:17:40	Saving to Database	OK

Report Generated by - MobSF v4.5.1

Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

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